

No. of Functional Features Included in the Solution

Date	15 March 2024
Team ID	UNM293SmartStudyAI
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	AI Story Script Generator	Generates multi-scene comic narrative scripts, dialogue bubbles, and scene descriptions using Gemini 1.5 Pro / Flash.	LLM Story Engine	Done	1.0
2	Character Persona & Consistency	Maintains visual style, outfit, and character features consistently across comic panels via engineered prompt conditioning.	Prompt Engineering & Style Module	Done	0.5
3	Panel Illustration Generator	Translates scene prompt scripts into stylized comic panels (manga, comic book, cartoon) using generative image synthesis.	Image Generation Pipeline	Done	1.0
4	Speech Bubble & Layout Editor	Interactive canvas editor enabling users to position captions, sound effects (onomatopoeia), and speech balloons on panels.	Canvas & Layout Builder	Done	0.5
5	Multi-Panel Comic Strip Builder	Supports dynamic comic strip grid templates (2x2, 3x1, 6-panel manga) with drag-and-drop panel reordering.	Frontend UI / Panel Grid	Done	0.5
6	Multi-Format Comic Exporter	Renders and exports completed comics into high-resolution printable PDF, CBZ archive, and PNG image sequences.	Export & Rendering Engine	Done	0.5
7	User Comic Project Management	Allows creators to save drafts, version their storylines, manage assets, and organize comic book series.	Backend & Database Module	Done	0.5
8	AI Comic Quality Critic & Enhancer	Uses Gemini Multimodal capabilities to evaluate visual flow, dialogue readability, and suggest pacing improvements.	Multimodal AI Evaluation	Done	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	5

Metric	Count / Value
Additional / Nice-to-Have Features	3
Features Tested & Verified	8 / 8 (100%)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Panel Grid Strip Builder, Speech Bubble Drag & Drop Editor
2	Backend / Logic	2	Project State Manager, High-Res PDF/CBZ Export Engine
3	Database / Storage	1	Comic Drafts, Scene Metadata & Character Asset Storage
4	API / Integration	2	Gemini 1.5 Pro Story Generation, Image Synthesis Pipeline
5	Security / Authentication	1	Secure API Key Management, User Auth & Session Control